

10/10  
Nicely done!

I.k.

All was used to proof the solution.

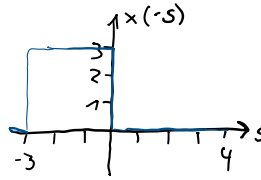
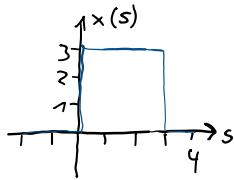
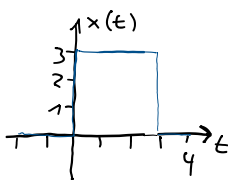
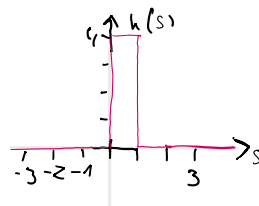
### 1. Graphical convolution (3 points)

The response  $A = x * h$  of a linear, time invariant, system is given by the convolution of the input signal  $x(t)$  and the system's impulse response  $h(t)$ . Here, we perform a "graphical" convolution between  $x$  and  $h$ , by computing the area under the product of the two functions. Consider the following functions:

$$x(t) = \begin{cases} 0 & \text{for } t < 0 \\ 3 & \text{for } 0 \leq t \leq 3 \\ 0 & \text{for } t > 3 \end{cases} \quad \text{and} \quad h(t) = \begin{cases} 0 & \text{for } t < 0 \\ 4 & \text{for } 0 \leq t \leq 1 \\ 0 & \text{for } t > 1 \end{cases}$$

(a) Sketch  $x(t-s)$  and  $h(s)$  as a function of  $s$  for a fixed, generic value of  $t$ . Identify the intervals of  $t$  for which the system has a different linear response. For each interval, sketch again  $x(t-s)$  and  $h(s)$  as a function of  $s$ . This will help you to compute the system's response.

(b) Compute the system's response  $A(t) = \int_{-\infty}^{\infty} x(t-s)h(s)ds$  and sketch the result.

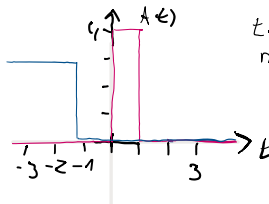


Edges of flipped signal  $\Rightarrow$  Shift by  $t$  to get  $x(t-s)$

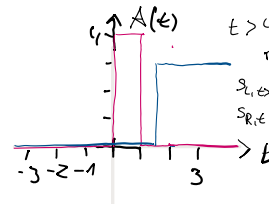
$$s_{L,0} = -3 \quad s_{R,0} = 0 \quad \Rightarrow \quad s_{L,t} = t + s_{L,0} = t - 3$$

$$s_{R,t} = t + s_{R,0} = t + 0 = t$$

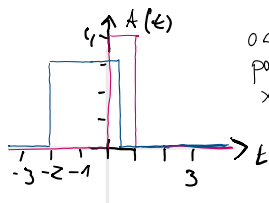
$t < 0$   
 $0 < t < 1$   
 $1 < t < 3$   
 $3 < t < 4$   
 $t > 4$



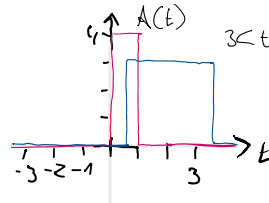
$t < 0$   
no overlap  
 $A(t) = 0$   
 $x(t-s)$   
 $s_{L,t} < -3$   
 $s_{R,t} < 0$



$t > 4$   
 $A(t) = 0$   
no overlap  
 $s_{L,t} > 1$   
 $s_{R,t} > 4$



$0 < t < 1$   
partly overlap  
 $x(t-s)$   
 $-3 < s_{L,t} < -2$   
 $0 < s_{R,t} < 1$



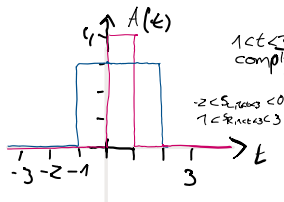
$3 < t < 4$   
partly overlap  
 $0 < s_{L,t} < 1$   
 $3 < s_{R,t} < 4$

Overlap depend on  $t$ .

Always overlap between  $t-3$  and  $1$ .  $s \in [t-3, 1]$

$$s \in [0, t] \\ A(t) = \int_0^t 3 \cdot 4 ds = 12[t-0] = 12t$$

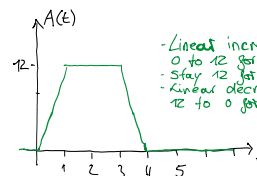
$$A(t) = \int_{t-3}^1 3 \cdot 4 ds = 12 \cdot [1 - (t-3)] = 12[4-t]$$



$1 < t < 3$   
complete overlap, intervals  $s \in [0, 1]$   
 $-2 < s_{L,t} < -1$   
 $1 < s_{R,t} < 3$

$$A(t) = \int_0^1 3 \cdot 4 ds = 12 \cdot [1-0] = 12$$

$$A(t) = \int_{-\infty}^{\infty} x(t-s)h(s)ds = \begin{cases} 0 & \text{for } t < 0 \\ 12t & \text{for } 0 < t < 1 \\ 12 & \text{for } 1 < t < 3 \\ 12(4-t) & \text{for } 3 < t < 4 \\ 0 & \text{for } t > 4 \end{cases}$$



- Linear increase from 0 to 12 for  $0 \leq t \leq 1$ .  
- Stay 12 for  $1 \leq t \leq 3$ .  
- Linear decrease from 12 to 0 for  $3 \leq t \leq 4$ .

3

## 2. Properties of the convolution (3 points)

Prove the following properties of the convolution:

- (i) Commutative:  $(u * v)(t) = (v * u)(t)$
- (ii) Distributive:  $(u * (\alpha v + \beta w))(t) = \alpha(u * v)(t) + \beta(u * w)(t)$  for all  $\alpha, \beta \in \mathbb{R}$
- (iii) Associative:  $(u * (v * w))(t) = ((u * v) * w)(t)$

where  $*$  stands for convolution. Hint: use the definition of the convolution and change of variables.

$$(u * v)(t) = \int_{-\infty}^{\infty} u(t-s) v(s) ds$$

i) commutative

$$(u * v)(t) = (v * u)(t)$$

$$\int_{-\infty}^{\infty} u(t-s) v(s) ds = \int_{-\infty}^{\infty} v(t-\tau) u(\tau) d\tau$$

$\tau = t-s$  and these implies  $s = t-\tau$  (change of variable)

$$\frac{ds}{d\tau} = \frac{d}{d\tau}(t-\tau) = -1 \Rightarrow ds = -d\tau$$

$$\text{With } s = t - \tau$$

$$\tau \rightarrow -\infty \text{ then } s \rightarrow \infty$$

$$\tau \rightarrow \infty \text{ then } s \rightarrow -\infty$$

$$\int_{-\infty}^{\infty} u(t-s) v(s) ds = \int_{\infty}^{-\infty} v(s) u(t-s) (-d\tau) = \int_{-\infty}^{\infty} v(s) u(t-s) ds$$

ii) distributive

$$(u * (\alpha v + \beta w))(t) = \alpha(u * v)(t) + \beta(u * w)(t)$$

Follows from linearity of integral

Left-hand side:

$$(u * (\alpha v + \beta w))(t) = \int_{-\infty}^{\infty} u(t-s) [\alpha v(s) + \beta w(s)] ds \quad \left| \begin{array}{l} \text{linearity} \\ \text{of integration} \end{array} \right.$$

$$= \alpha \int_{-\infty}^{\infty} u(t-s) v(s) ds + \beta \int_{-\infty}^{\infty} u(t-s) w(s) ds$$

$$= \alpha \cdot (u * v) + \beta \cdot (u * w)$$

iii) Associative

$$(u * (v * w))(t) = ((u * v) * w)(t)$$

Follows Fubini's Theorem, i.e. double integrals

$$\text{can be evaluated as iterated integrals in either order: } \iint_{\mathbb{R}^2} f(x,y) dx dy = \int_{\mathbb{R}} \left( \int_{\mathbb{R}} f(x,y) dx \right) dy = \int_{\mathbb{R}} \left( \int_{\mathbb{R}} f(x,y) dy \right) dx$$

$$(u * v) * w(t) = \int_{-\infty}^{\infty} (u * v)(t-s) w(s) ds$$

$$= \int_{-\infty}^{\infty} \left[ \int_{-\infty}^{\infty} u(t-s-\tau) v(\tau) d\tau \right] w(s) ds$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(t-s-\tau) v(\tau) w(s) d\tau ds$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(t-s-\tau) v(\tau) w(s) ds d\tau \quad \text{commutative}$$

$$= \int_{-\infty}^{\infty} u * w(t-\tau) v(\tau) d\tau = (u * w) * v = u * (v * w)$$

## 3. The Dirac delta (4 points)

The Dirac delta is commonly used to analytically describe the spike times of a neuron. This exercise gets you acquainted with two ways of defining the Dirac delta and how one of them is not mathematically rigorous.

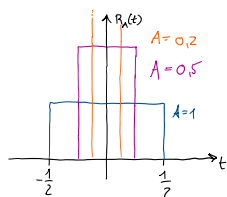
- (a) Consider a rectangular function  $R_A(t)$  centered at 0 with width  $A$  and height  $1/A$  such that

$$R_A(t) = \begin{cases} 1/A & \text{for } -\frac{A}{2} \leq t \leq \frac{A}{2} \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

- (i) Sketch and explain what happens to  $R_A$  for arbitrarily small  $A$ .

- (ii) Calculate  $\int_{-\infty}^{\infty} R_A(t) dt$  for any  $A > 0$ .

i)



Area under the curves stays the same, as Area = width  $\cdot$  height =  $A \cdot \frac{1}{A} = 1$

$$A=1 \quad w=1 \quad h=1$$

$$A=0.5 \quad w=0.5 \quad h=2$$

$$A=0.2 \quad w=0.2 \quad h=2.5$$

For  $A \rightarrow 0$   $R_A$  gets smaller and width and taller in height. So  $R_A$  becomes a narrow pulse, "converges" to Dirac delta  $\delta(t)$  which is infinity at  $t=0$  and 0 elsewhere.

Area under the curve says the same, as Area = width · height =  $A \cdot \frac{1}{A} = 1$

height. So  $R_A$  becomes a narrow pulse, "converges" to Dirac delta  $\delta(t)$  which is infinity at  $t=0$  and 0 elsewhere.  
 $\int_{-\infty}^{\infty} \delta(t) dt = 1$

$$1) \int_{-\infty}^{\infty} R_A(t) dt = \int_{-\frac{A}{2}}^{\frac{A}{2}} R_A(t) dt$$

As  $R_A(t) = 0$  outside  $[-\frac{A}{2}, \frac{A}{2}]$

$$\int_{-\frac{A}{2}}^{\frac{A}{2}} \frac{1}{A} dt = \frac{1}{A} \left[ \frac{A}{2} - \left(-\frac{A}{2}\right) \right] = \frac{1}{A} \cdot A = 1$$

(c) Let's now look at a mathematically correct definition: The Dirac delta  $\delta(t)$  has the defining property

$$\int_{-\infty}^{\infty} f(t-s) \delta(s) ds = f(t). \quad (3)$$

$\delta(t)$  is not a function in the common sense but a distribution (or "generalized function"). Thus, the left hand side of equation (3) is not a Riemann integral but can instead be defined as:

$$\int_{-\infty}^{\infty} f(t-s) \delta(s) ds := \lim_{A \rightarrow 0} \int_{-\infty}^{\infty} f(t-s) R_A(s) ds. \quad (4)$$

(i) Show that  $\int_{-\infty}^{\infty} f(t-s) R_A(s) ds = \frac{F(t+A/2) - F(t-A/2)}{A}$ , where  $F$  is a primitive function of  $f$ .

(ii) Show that the defining property (3) of the Dirac delta holds for (4).

Hint: Use (i) and the definition of derivative  $\frac{d}{dt} f(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t-h)}{2h}$ .

(iii) Use (3) to show that

$$\int_{-\infty}^{\infty} \delta(s) ds = 1. \quad (5)$$

Keep in mind that the Dirac delta is not a function but a distribution. Even though it is often intuitively also used on its own, it is an object that rigorously only makes sense when applied to a so-called test function  $f$  (see (3)).

c)

$$1) \int_{-\infty}^{\infty} f(t-s) R_A(s) ds$$

$$= \int_{-\frac{A}{2}}^{\frac{A}{2}} f(t-s) R_A(s) ds \quad \left| \begin{array}{l} R_A(s) \text{ is 0} \\ \text{outside } [-\frac{A}{2}, \frac{A}{2}] \end{array} \right.$$

$$= \frac{1}{A} \int_{-\frac{A}{2}}^{\frac{A}{2}} f(t-s) ds \quad \left| \begin{array}{l} \text{substitute } u = t-s \\ s = t-u \quad ds = -du \end{array} \right.$$

$$= \frac{1}{A} \int_{t+\frac{A}{2}}^{t-\frac{A}{2}} f(u) (-du) = \frac{1}{A} \int_{t-\frac{A}{2}}^{t+\frac{A}{2}} f(u) du \quad \left| \begin{array}{l} \text{we know:} \\ \int_a^b f(u) du = F(b) - F(a) \end{array} \right.$$

$$= \frac{1}{A} \left[ F\left(t+\frac{A}{2}\right) - F\left(t-\frac{A}{2}\right) \right] = \frac{F\left(t+\frac{A}{2}\right) - F\left(t-\frac{A}{2}\right)}{A}$$

ii) Proof that

$$\lim_{A \rightarrow 0} \int_{-\infty}^{\infty} f(t-s) R_A(s) ds = f(t) \quad \left| \text{use i)} \right.$$

$$\lim_{A \rightarrow 0} \frac{F\left(t+\frac{A}{2}\right) - F\left(t-\frac{A}{2}\right)}{A} \quad \left| \begin{array}{l} \text{substitute} \\ h = \frac{A}{2} \end{array} \right.$$

$$\lim_{h \rightarrow 0} \frac{F(t+h) - F(t-h)}{2h} \quad \left| \begin{array}{l} \text{symmetric} \\ \text{difference} \\ \text{quotient} \end{array} \right.$$

$$= F'(t) = f(t)$$

$$11) \int_{-\infty}^{\infty} f(t-s) \delta(s) ds = f(t)$$

Test function  $f(x) = 1 \forall x$

Then  $f(t-s) = 1$  and  $f(t) = 1$  (because  $f$  is constant)

$$\int_{-\infty}^{\infty} 1 \cdot \delta(s) ds = 1$$